

EDUMATCH

Predictive Modeling and Prescriptive Analytics for Early Student Retention Intervention

An end-to-end educational performance metric with local regulatory knowledge bases using machine learning and RAG architecture.

TECH STACK: Python, Pandas, NumPy, Scikit-Learn (Random Forest, K-Means, Standard Scaler), Matplotlib, Seaborn, LLMS, Streamlit

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Main Objective: Mitigating Early Student Attrition



The Core Problem

Higher education institutions face severe drop-out rates during the first two semesters of undergraduate study.



Systemic Challenge

Traditional intervention models react to student failures — registration anomalies or failing grades — rather than acting preemptively.

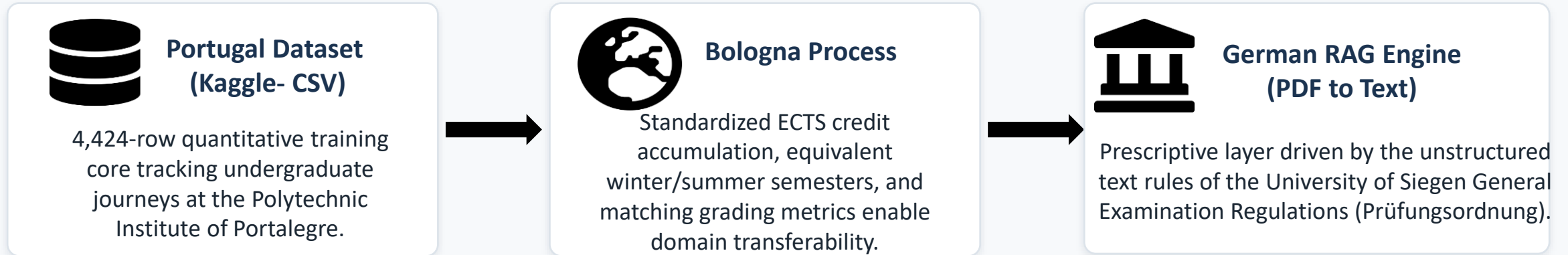


Action Pathway

This project constructs an end-to-end predictive pipeline to identify ‘at-risk’ students from baseline demographics, socioeconomic stressors, and initial academic trajectory.

The Database: Ingestion and Exploration

Data Alignment: Portugal Dataset to German RAG Engine



Data Harmonization Strategy

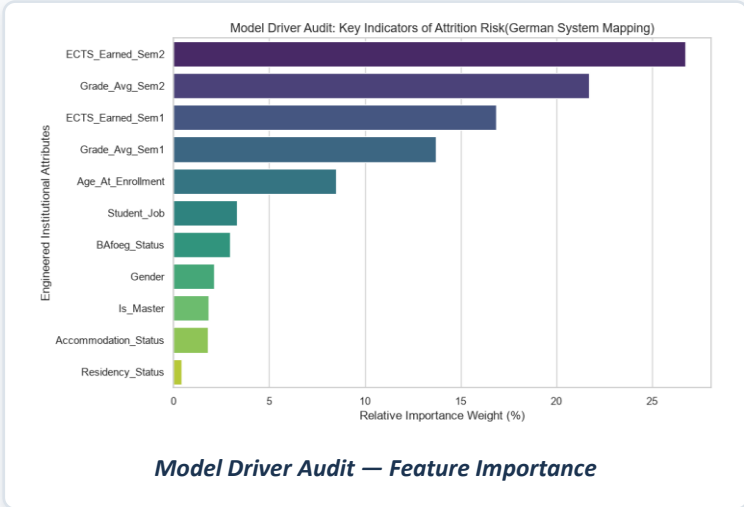
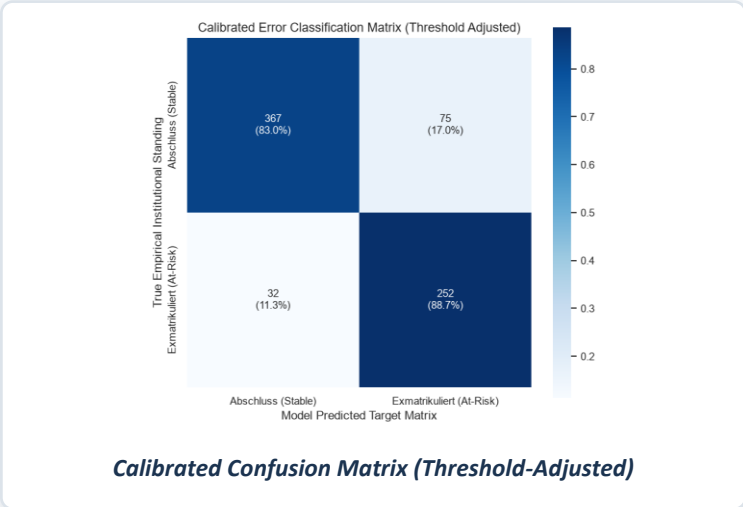
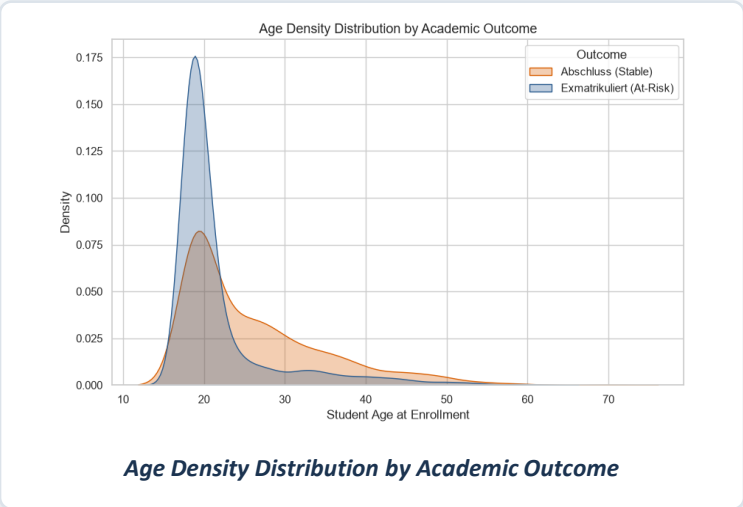
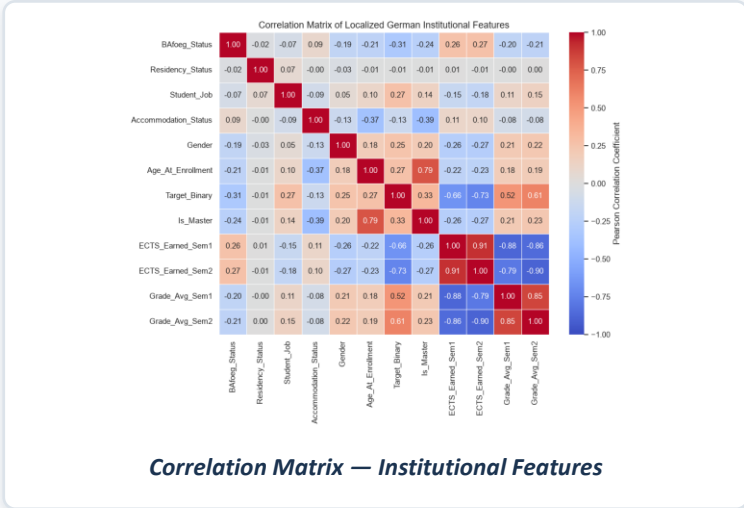
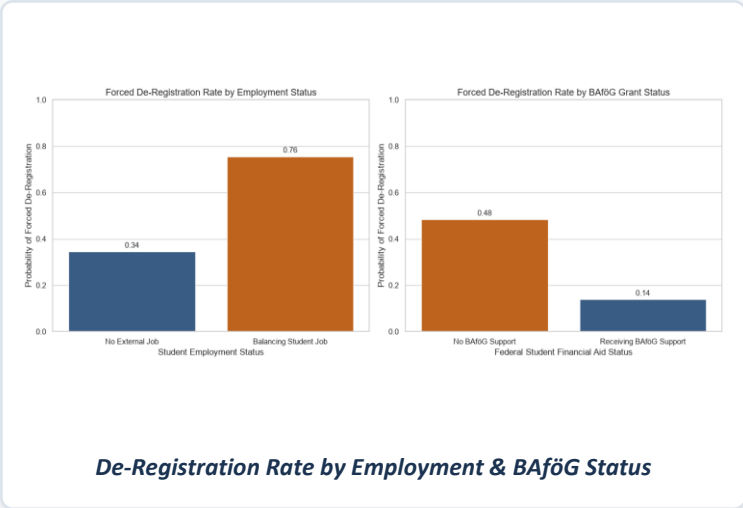
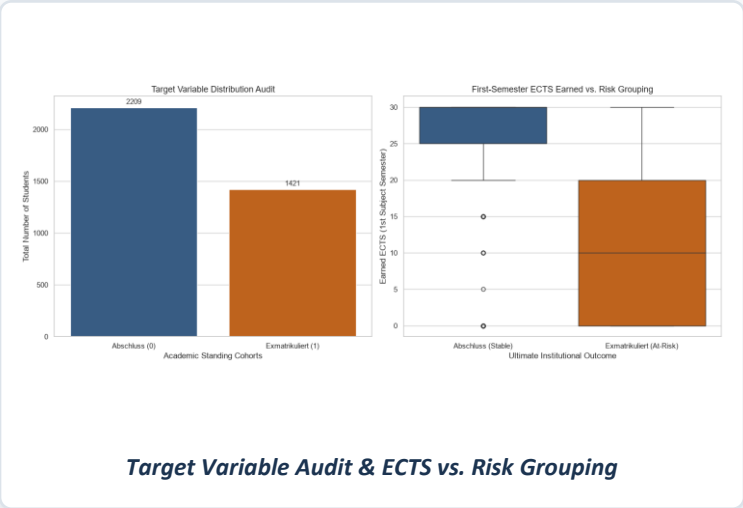
Though trained on Portuguese behavioral features and applied to German rules, both institutions operate under the European Bologna Process.

Unified ECTS benchmarks and matching semester structures allow robust domain transferability between the two academic systems.

Unified Features: ECTS credit accumulation • equivalent winter/summer semester divisions • matching baseline grading metrics

Supporting Data Visualization Images

EDA, model driver, and validation charts underpinning the EduMatch pipeline



Decision Boundary Re-Calibration and K-Means Clustering

Strategic Threshold Tuning



Set a manageable margin of precision to reduce the number of missed at-risk individuals for early intervention when warning indicators appear.

Unsupervised Persona Profiling — K-Means Clustering of At-Risk Students



Acute Academic Collapse

Severe, fast-moving academic breakdown – requiring immediate peer-mentorship intervention.



Stabilized Academic Deficit

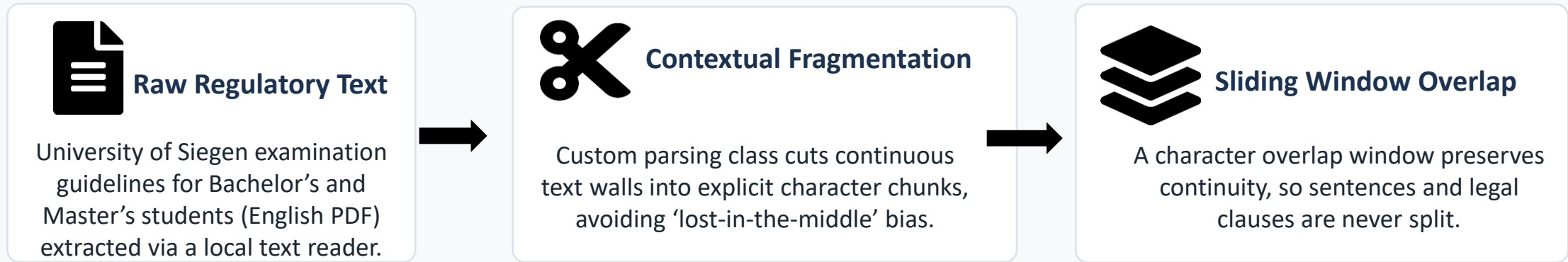
Persistent but manageable underperformance –routed to targeted academic advisory.



Latent Socioeconomic Risk

Financial stress, residential status, and accommodation stability as socio-economic impact factors –needing proactive intervention

Knowledge Base Extraction and Chunking



Local Knowledge Assembly: All outputs are consolidated into a portable local staging index (`raw_extracted_po.txt`), bypassing external cloud dependencies.

Streamlit UI Interactive Dashboard

 EduMatch: Predictive Student Retention and Prescriptive Analytics

 Advisor Input Panel

Age at Enrollment

22

Gender

Female

Enrolled Degree Level

Bachelor Level Degree Program

Residency Classification

EU / Domestic Student

BARSG Recipient Status

No

Employment Configuration

No Job (Full-Time Study Focus)

Accommodation Stability

Stable Housing Structure

Academic Milestones

ECTS Credits Earned (Sem 1)

30

Grade Average (Sem 1) [1.0 Best to 5.0 Fail]

1.78

ECTS Credits Earned (Sem 2)

30

Grade Average (Sem 2) [1.0 Best to 5.0 Fail]

2.43

Run Prediction & RAG Analysis

Clear All Inputs

 Ad-Hoc Regulatory Consultation Sandbox

Ask an ad hoc, free text question regarding specific regulations or edge cases for this student:

Enter your specific advisor question here:

Query Examination Database

 Live Analytics Engine

 **HIGH RETENTION ALERT: 89.0% Attrition Probability (Threshold: 35.0%)**

 Risk Driver Deconstruction

Operational Academic Risk Score

21.0%

Progress metrics stable.

Socioeconomic & Environmental Strain

45.0%

Structural context clear.

Assigned Support Intervention Cohort

Cohort Focus: Cluster 3: Employed Student Study Work Pressure Profile

 Vector-Matched Examination Regulations (Prüfungsordnung)

 Academic Advisory Assessment Report

Academic Standing

- Semester 1: Gut (Grade 1.7), indicating a good academic performance.
- Semester 2: Gut (Grade 2.4), still within the good performance range but showing a decline from Semester 1.

Trend Analysis

- The increase in grade number from 1.7 to 2.4 indicates a steep performance decline, sounding a warning flag for the student's academic progress.
- The student has completed 1 year of progress with 60 ECTS, which is a significant milestone.

Regulatory Directives

- Ensure timely payment of fees prior to semester deadlines to maintain registration validity, as per the General Examination Regulations Framework [PO-101].
- Monitor and address the worsening performance trend to prevent further decline, considering the high attrition risk score of 89.0%.
- Develop strategies to improve academic performance, focusing on reversing the current worsening gradient direction to achieve better grades, ideally within the Sehr Gut (1.0-1.5) or at least maintaining a Gut (1.6-2.5) range.

>  View Source Clauses [PO-101]

THANK YOU FOR YOUR ATTENTION

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ME DAASE!

**I'm happy to take questions, feedback, or recommendations
for my next career.**